Year 11 Mathematics

Shape and Space 5

5.1 Circle Theorems

Circle Theorems

STARTER · 5 MIN

Date: _____ **Title:** Circle Theorems

LEARNING OBJECTIVES

- Use centre, semicircle and same segment theorems
- Use cyclic quadrilateral and tangent theorems
- Use the alternate segment theorem and give reasons

STARTER *Answer in your book. No calculators.*

1. Angles in a triangle add up to?
2. Find the third angle: 50° and 60°
3. Isosceles triangle, top angle 40°. Find a base angle.
4. Angles on a straight line: 125° and x

Starter — Answers

STARTER

Tick or correct in green pen

1. **180°**

2. $180 - 110 = \mathbf{70°}$

3. $(180 - 40) \div 2 = \mathbf{70°}$

4. $180 - 125 = \mathbf{55°}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can use centre, semicircle and same segment theorems	☐
SILVER	I can use cyclic quadrilateral and tangent theorems	☐
GOLD	I can use the alternate segment theorem and give reasons	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Circle Theorems

INTRODUCTION · 3 MIN

Circle theorems give angle facts. Always state the theorem as your reason.

Centre

Angle at centre = 2 × angle at circumference.

Circumference 50° → centre 100°

Semicircle and segment

Angle in a semicircle is 90°. Angles in the same segment are equal.

Diameter → 90°

Cyclic quadrilateral

Opposite angles add up to 180°.

75° + x = 180°

Tangents

Tangent ⊥ radius. Tangents from a point are equal.

Alternate segment theorem

Examples 1 and 2 — Centre, Semicircle and Same Segment

EXAMPLES · 5 MIN

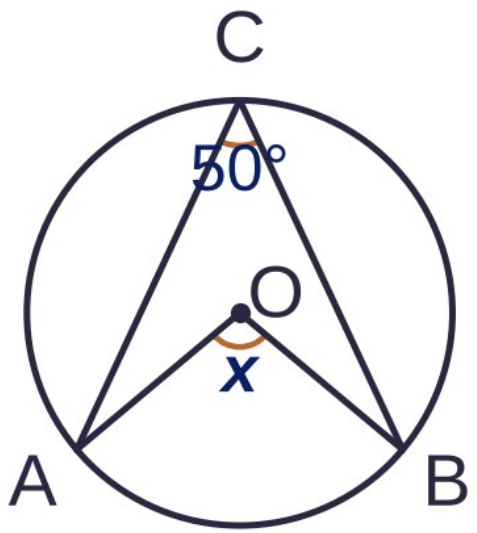
BRONZE

Watch and copy into your book

Hint centre = 2 × circumference semicircle = 90°

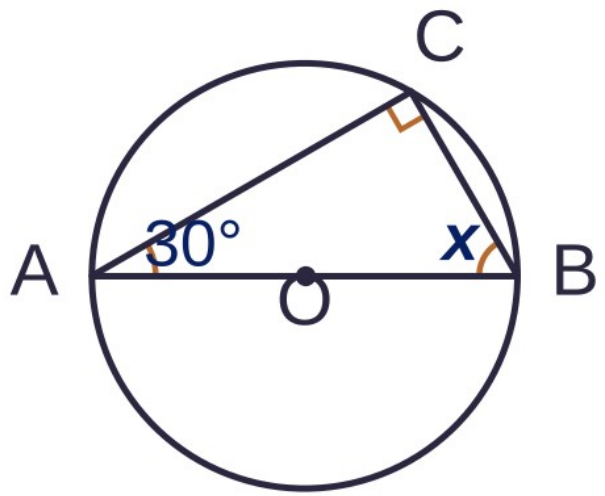
Example 1

O is the centre. Find x.



Example 2

AB is a diameter. Find x.



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Centre angle is double

$x = 2 \times 50$

$x = 100°$

Tip: the centre angle is double.

Example 2

Semicircle angle is 90°

$x = 180 - 90 - 30$

$x = 60°$

Tip: look for the diameter.

Task 1

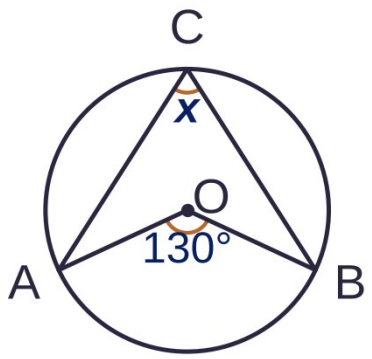
PRACTICE · 9 MIN

BRONZE

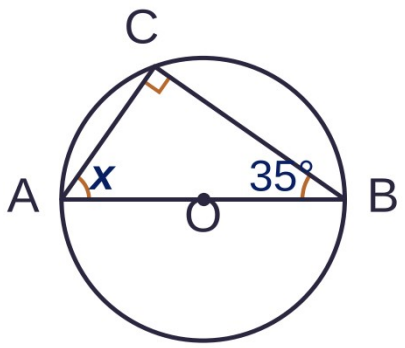
O is the centre. Find x.

Hint centre = 2 × circumference semicircle = 90°

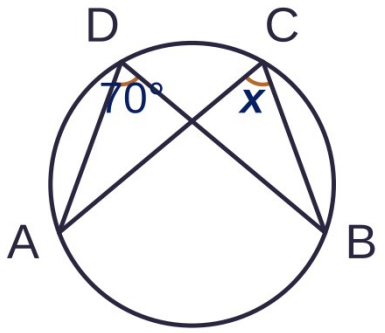
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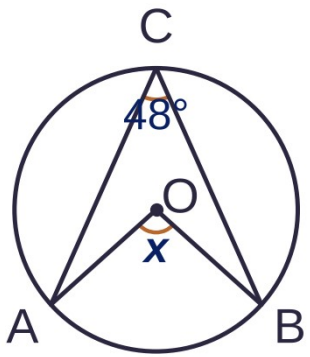
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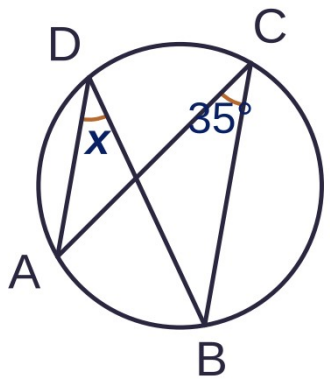
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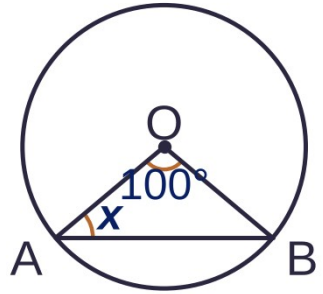
4



5



★



Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $130 \div 2 = \mathbf{65°}$

2. $180 - 90 - 35 = \mathbf{55°}$

3. Same segment, $x = \mathbf{70°}$

4. $48 \times 2 = \mathbf{96°}$

5. Same segment, $x = \mathbf{35°}$

★ $(180 - 100) \div 2 = \mathbf{40°}$

Examples 3 and 4 — Cyclic Quadrilaterals and Tangents

EXAMPLES · 5 MIN

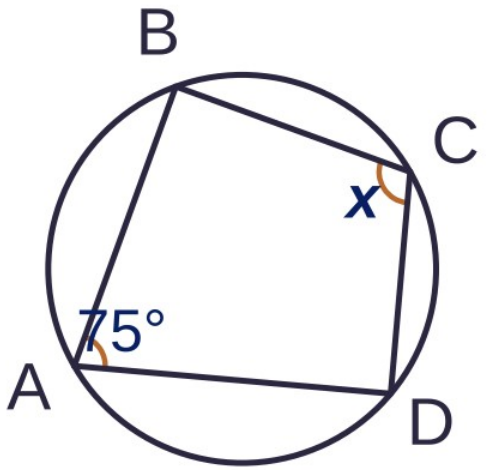
SILVER

Watch and copy into your book

Hint cyclic quad: opposite angles = 180° tangent ⊥ radius

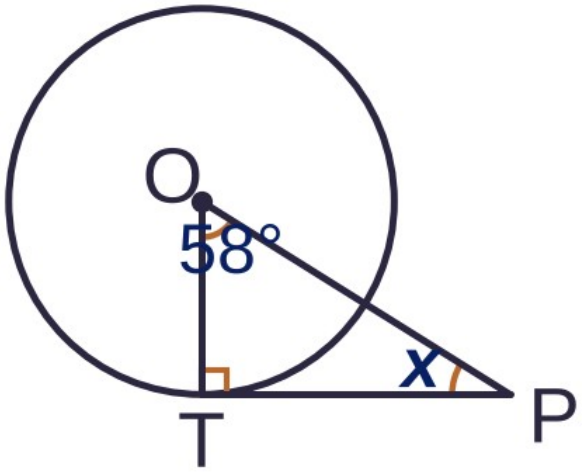
Example 3

ABCD is a cyclic quadrilateral. Find x.



Example 4

PT is a tangent. O is the centre. Find x.



Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Opposite angles add to 180°

$x = 180 - 75$

$x = 105°$

Tip: opposite, not next to each other.

Example 4

Tangent ⊥ radius

$x = 180 - 90 - 58$

$x = 32°$

Tip: mark the right angle first.

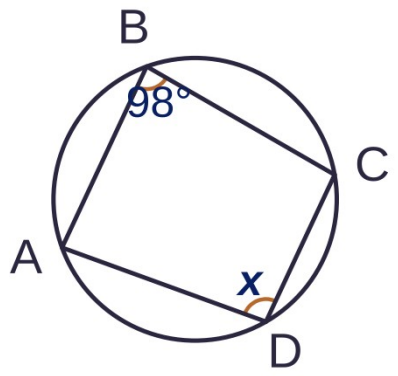
Task 2

PRACTICE · 9 MIN

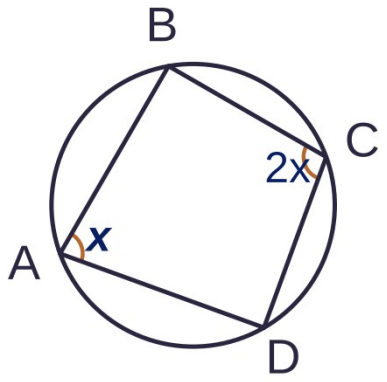
SILVER

Find x. O is the centre. PA, PB and PT are tangents.

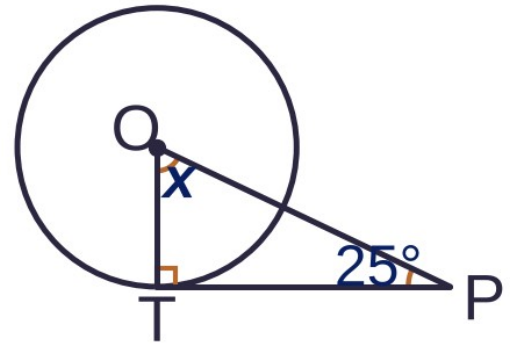
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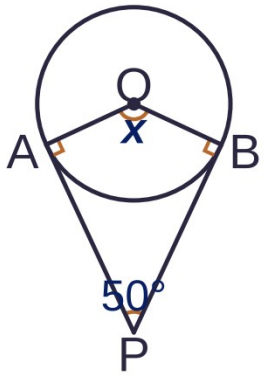
2



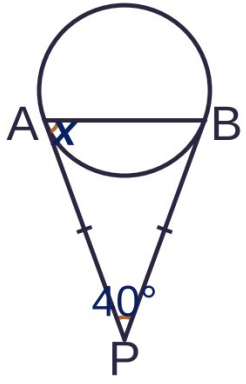
3



4



5



Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $180 - 98 = \mathbf{82°}$

2. $3x = 180, \quad x = \mathbf{60°}$

3. $180 - 90 - 25 = \mathbf{65°}$

4. $360 - 90 - 90 - 50 = \mathbf{130°}$

5. $(180 - 40) \div 2 = \mathbf{70°}$

Examples 5 and 6 — Alternate Segment and Reasons

EXAMPLES · 5 MIN

GOLD

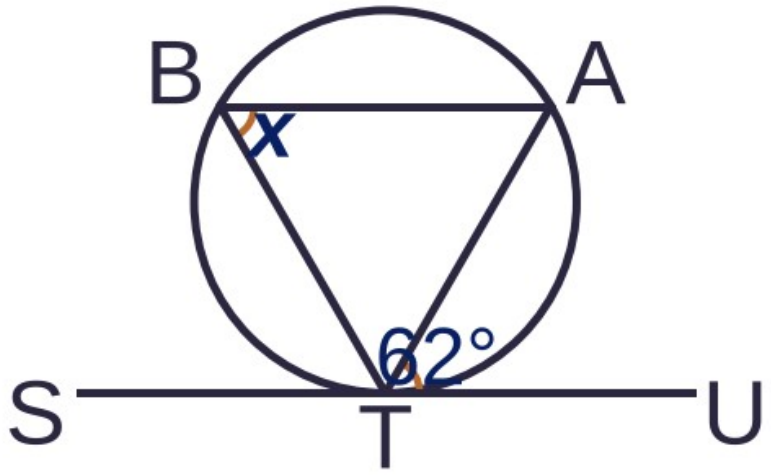
Exam-style: show all your working

Hint tangent–chord angle = angle in the alternate segment

Example 5

(2 marks)

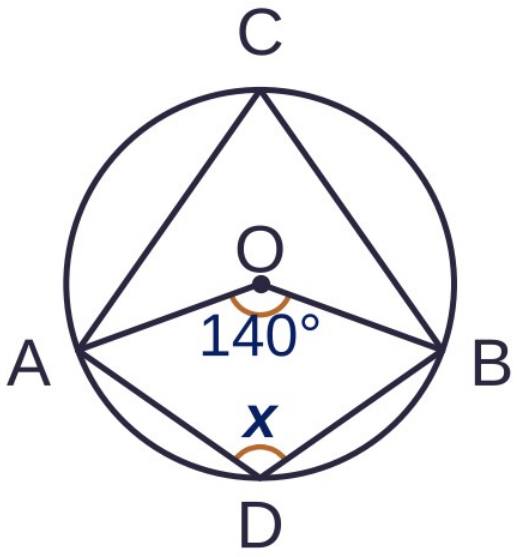
SU is a tangent at T. Find x. Give a reason.



Example 6

(3 marks)

O is the centre. Find x. Give reasons.



Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Alternate segment theorem

x = 62°

Tip: the angle is in the opposite segment.

Example 6

ACB = 140 ÷ 2

= 70° (centre)

x = 180 − 70 (cyclic quad)

x = 110°

Tip: find angle ACB first.

Task 3

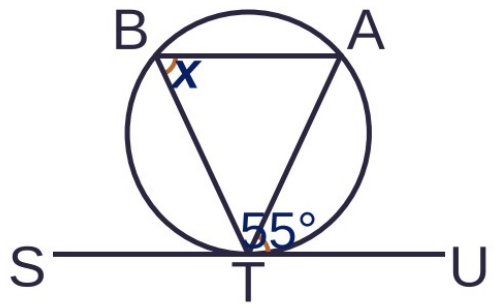
PRACTICE · 9 MIN

GOLD

Find the angles. Give reasons.

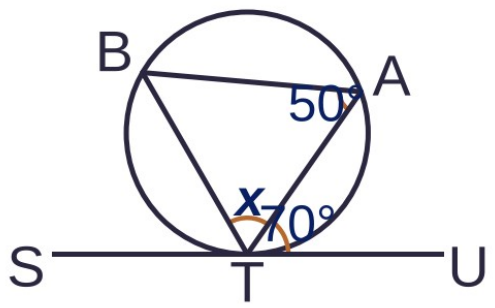
Hint tangent–chord angle = angle in the alternate segment

1



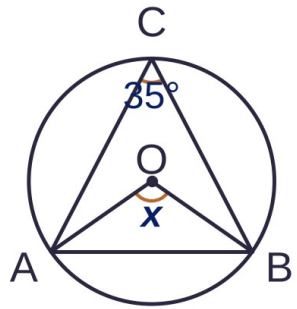
(2 marks)

2



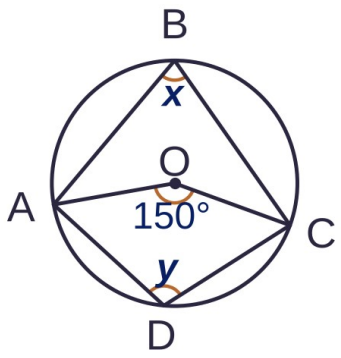
(3 marks)

3



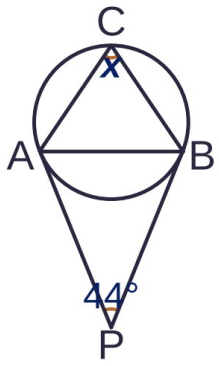
(3 marks)

4



(3 marks)

5



(3 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $x = \mathbf{55°}$ **(alternate segment)**

2. $B = 70°, \quad x = 180 - 70 - 50 = \mathbf{60°}$

3. $x = 35 \times 2 = \mathbf{70°}$ **(centre)**

4. $x = 150 \div 2 = 75°, \quad y = 180 - 75 = \mathbf{105°}$

5. $PAB = (180 - 44) \div 2 = 68°, \quad x = \mathbf{68°}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE The angle at the centre is 110°. Find the angle at the circumference.

SILVER One angle of a cyclic quadrilateral is 64°. Find the opposite angle.

GOLD A tangent–chord angle is 48°. Find the angle in the alternate segment.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $110 \div 2 = \mathbf{55°}$

SILVER $180 - 64 = \mathbf{116°}$

GOLD **48°**

Exam Practice — Centre and Semicircle

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint centre = 2 × circumference semicircle = 90°

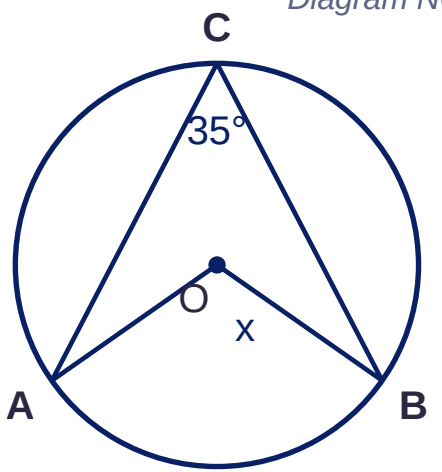
Question 1

(2 marks)

A, B and C are points on the circle, centre O.

Find x. Give a reason.

Diagram NOT accurately drawn



x = ...

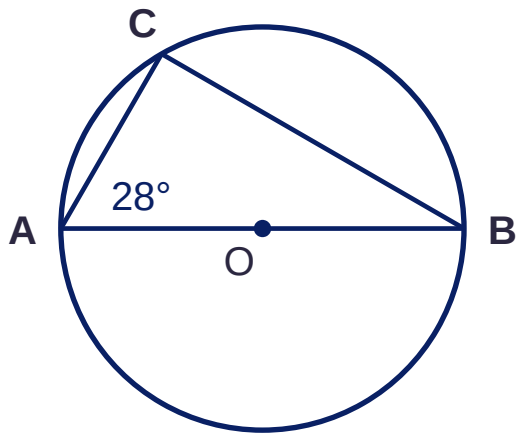
Question 2

(2 marks)

AB is a diameter of the circle, centre O.

Find angle ABC. Give reasons.

Diagram NOT accurately drawn



angle ABC = ...

Exam Practice — Cyclic Quadrilaterals and Tangents

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint cyclic quad: opposite angles = 180° tangent ⊥ radius

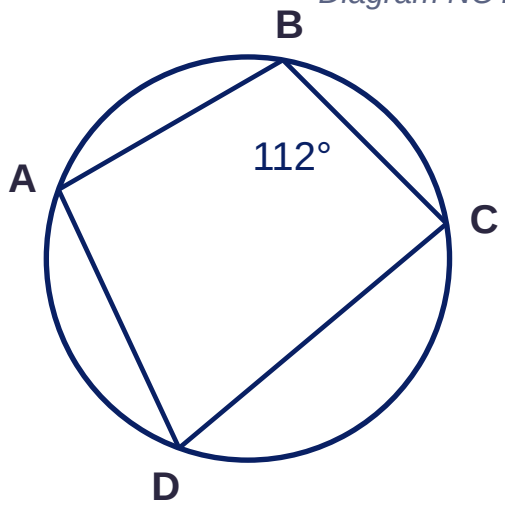
Question 3

(2 marks)

ABCD is a cyclic quadrilateral.

Find angle ADC. Give a reason.

Diagram NOT accurately drawn



angle ADC = ...

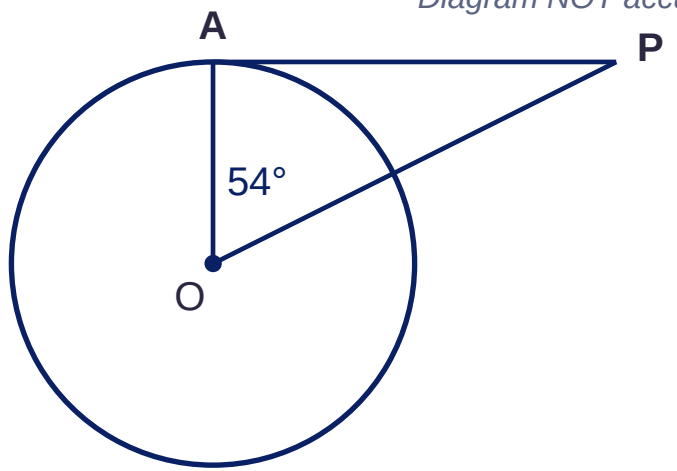
Question 4

(2 marks)

PA is a tangent to the circle at A.

O is the centre. Find angle APO.

Diagram NOT accurately drawn



angle APO = ...

Exam Practice — Alternate Segment and Reasons

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

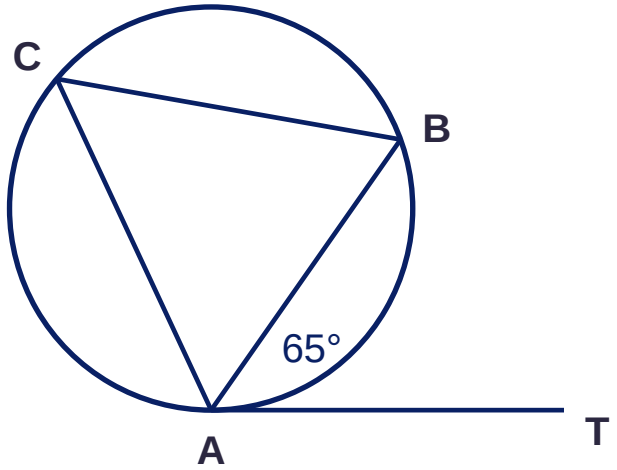
Hint tangent–chord angle = angle in the alternate segment

Question 5

(2 marks)

TA is a tangent at A. Find angle ACB.
Give a reason.

Diagram NOT accurately drawn



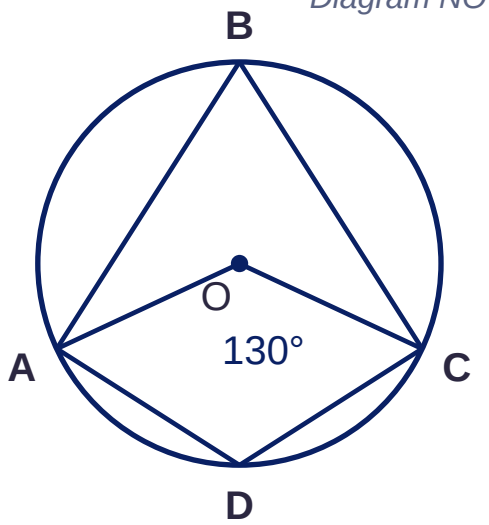
angle ACB = ...

Question 6

(3 marks)

A, B, C, D lie on a circle, centre O.
Find angle ADC. Give reasons.

Diagram NOT accurately drawn



angle ADC = ...

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

x = 2 × 35 = 70° B1

angle at the centre is twice B1

the angle at the circumference

Tip: the centre angle is double.

Question 2

angle ACB = 90° B1

angle in a semicircle is 90°

ABC = 180 – 90 – 28 = 62° B1

Tip: look for the diameter.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

180 − 112 — M1

= 68° — A1

opposite angles of a cyclic

quadrilateral add up to 180°

Tip: opposite, not next to each other.

Question 4

angle OAP = 90° — M1

tangent ⊥ radius

APO = 180 – 90 – 54 = 36° — A1

Tip: mark the right angle first.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

angle ACB = 65° — B1

alternate segment theorem — B1

Tip: the angle is in the opposite segment.

Question 6

ABC = 130 ÷ 2 = 65° — M1

centre angle is double

ADC = 180 – 65 = 115° — A1

cyclic quadrilateral — B1

Tip: find angle ABC first.